Federated Learning and Over-the-Air Computation: A Comparative Study

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***Abstract*—As Federated Learning (FL) emerges as a privacy-preserving alternative to centralized machine learning, ensuring communication efficiency becomes increasingly crucial. This paper examines two promising strategies: orthogonal FL utilizing sequential transmissions and Over-the-Air FL, where signals are combined in the air simultaneously. Their performance is evaluated across several key parameters to uncover trade-offs between accuracy and resource usage. A simulation framework was designed to assess convergence behavior under varying dataset complexities and data distributions. While orthogonal FL achieves higher accuracy and greater stability, Over-the-Air FL offers substantial communication savings, particularly in noise-resilient scenarios. These findings highlight important trade-offs between model performance and communication efficiency.**

***Sammanfattning*—I takt med att Federated Learning (FL) växer fram som ett integritetssäkert alternativ till centraliserad maskininlärning blir kommunikationseffektivitet allt viktigare. Denna rapport undersöker två lovande strategier: ortogonal FL med sekventiella överföringar samt Over-the-Air FL, där signaler kombineras direkt i luften genom samtidig överföring. Deras prestanda utvärderas utifrån flera centrala parametrar för att identifiera avvägningar mellan noggrannhet och resursanvändning. Ett simuleringsramverk har utformats för att analysera konvergensbeteende vid varierande datakomplexitet och datadistributioner. Även om ortogonal FL uppnår högre noggrannhet och större stabilitet, erbjuder Over-the-Air FL betydande kommunikationsvinster, särskilt i brusresistenta scenarier. Resultaten belyser viktiga avvägningar mellan modellprestanda och kommunikationseffektivitet.**

***Index Terms*—Machine learning, Federated Learning, Over-the-Air Computation.**

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I. Introduction

Neural Networks form the foundation of machine learning models, enabling pattern recognition by mimicking the structure and function of biological neurons [1]. Traditionally, these models are trained in a centralized manner, which raises privacy concerns. Federated Learning (FL) addresses this issue by enabling distributed model training directly on user devices, where only model updates are shared instead of raw data [2].

Orthogonal FL consists of orthogonal communication and the digital aggregation method Federated Averaging (FedAvg). FedAvg aggregates locally computed model updates on a central server by averaging them. When deployed in wireless networks, FedAvg typically relies on orthogonal communication, where each client is allocated a separate time or frequency resource to avoid interference [3]. Orthogonal FL ensures reliable aggregation and stable convergence, but it also leads to increased communication overhead, as the required bandwidth scales linearly with the number of clients [2].

Over-the-Air Computation (AirComp) has emerged as a promising alternative through its use of non-orthogonal communication, allowing multiple clients to transmit simultaneously over the same channel. By exploiting the superposition property of wireless signals, AirComp performs aggregation during transmission, significantly reducing communication latency and bandwidth usage [4]. However, this efficiency comes at the cost of increased sensitivity to noise, interference, and synchronization errors. When AirComp is combined with FL, the resulting paradigm is commonly referred to as Over-the-Air FL. This framework leverages analog signal superposition to approximate the aggregation step in FL, aiming to replicate the effect of FedAvg while drastically reducing communication demands [2].

Recent research has focused on improving the efficiency of both orthogonal FL and non-orthogonal Over-the-Air FL by optimizing model aggregation and reducing communication overhead. In orthogonal FL, enhancements include reducing the number of communication rounds and improving convergence on heterogeneous data [2]. In Over-the-Air FL, techniques such as threshold-based client selection and multi-receiver architectures have been developed to mitigate the effects of channel distortion and noise [4]. These advancements motivate a comparative evaluation of orthogonal and non-orthogonal FL systems under varying conditions.

The project is subject to several limitations to ensure feasibility and focus. The study is restricted to two datasets, which define the scope of evaluation. Furthermore, the implementations of orthogonal FL and Over-the-Air FL prioritize pedagogical clarity over state-of-the-art performance. Lastly, time constraints limit the number and complexity of experiments conducted.

Building on these considerations, this project investigates FL in two distinct communication scenarios: orthogonal FL, using an orthogonal communication based on FedAvg, and Over-the-Air FL, using a non-orthogonal communication utilizing AirComp. The aim is to evaluate and compare these approaches across key performance metrics, such as convergence behavior, accuracy, and communication efficiency. Through this analysis, the study highlights the trade-offs associated with each communication strategy and provides insights into their practical implications. In doing so, the thesis contributes to a deeper understanding of the broader interplay between orthogonal and non-orthogonal methods in FL.

II. Theoretical Background

This section outlines the theoretical foundations necessary to understand the comparison between Federated Learning (FL) under different communication paradigms. The section begins with a brief overview of neural network architectures, datasets, and data distributions. The general theory of FL is then introduced, followed by an explanation of the communication and aggregation of orthogonal FL with the Federated Averaging (FedAvg) algorithm. Finally, the concept of Over-the-Air FL is presented by explaining how Over-the-Air Computation (AirComp) approximates FedAvg in a non-orthogonal communication setting.

A. Neural Networks and Data Representation

Neural networks are computational models inspired by biological neural structures, particularly the human brain. These networks consist of interconnected layers of artificial neurons, where each neuron applies a transformation to its inputs using a set of learned parameters. The learning process is driven by gradient-based optimization methods to minimize a loss function, a measure of the discrepancy between the network's predictions and the actual target values [1].

This study employs two neural network architectures: Multilayer Perceptron (MLP) and Convolutional Neural Network (CNN). The MLP is a fully connected network with hidden layers that apply non-linear transformations to input data, making it effective for simple classification tasks. In contrast, the CNN is a deep learning model particularly well-suited for more complex classification tasks than MLP, as it utilizes convolutional layers to extract spatial features from images [5].

This study utilizes two datasets, MNIST and CIFAR-10. The MNIST dataset consists of handwritten digits ranging from zero to nine, with 60,000 training images and 10,000 test images. Each image is grayscale and measures 28×28 pixels [6]. In comparison, CIFAR-10 contains 60,000 color images measuring 32×32 pixels, divided into 50,000 training images and 10,000 test images across 10 object categories [7].

FL scenarios are commonly studied under two main types of data distributions: Independent and Identically Distributed (IID) and non-Independent and Identically Distributed (non-IID). In the IID setting, each device receives a subset of data that reflects the overall distribution of the dataset. In contrast, the non-IID setting assigns data in a way that is biased toward specific categories, causing individual device datasets to diverge from the global distribution. This imbalance poses challenges for distributed learning, as local updates may vary significantly, making it more difficult for the global model to converge effectively [8].

B. Fundamentals of Federated Learning

FL is a decentralized machine learning paradigm that enables multiple devices, also referred to as clients, to collaboratively train a global model without sharing their raw data. Instead of transferring private data to a central server, each client trains a local model and communicates only the learned updates, such as model weights or gradients, to a global server, as illustrated in Fig. 1. This approach preserves user privacy and reduces the risk of data leakage, while still allowing the global model to benefit from distributed datasets [2].

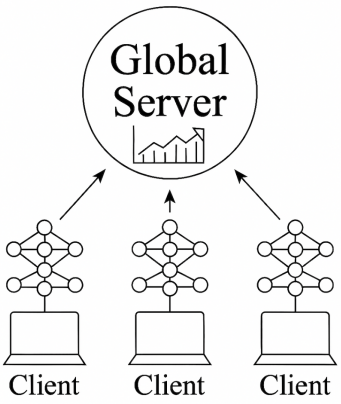


Fig. 1. Illustration of the FL process.

The primary motivation for FL is the increasing demand for privacy-preserving machine learning applications in sectors such as healthcare, finance, and mobile device personalization. Traditional centralized training approaches require collecting data in one location, which is often infeasible due to privacy regulations. FL mitigates this challenge by keeping the data localized and transmitting only client updates [2].

The primary objective of FL is to minimize the global loss function by aggregating local updates from multiple clients. This global loss function is defined as a weighted sum of local loss functions across all participating clients:

$$\min_{w} f(w) = \sum_{k=1}^{K} \frac{n_k}{n} F_k(w), \tag{1}$$

where w represents the model parameters, K denotes the total number of clients, and n_k corresponds to the number of data points held by client k. The total number of data points across all clients is given by $n = \sum_{k=1}^{K} n_k$. Each client minimizes its local loss function:

$$F_k(w) = \frac{1}{n_k} \sum_{i \in \mathcal{D}_k} \ell(w; x_i, y_i), \tag{2}$$

where $\ell(w; x_i, y_i)$ represents the loss function computed for each training sample (x_i, y_i) [2]. Here, x_i is a vector representing the image and y_i is the corresponding label verifying what x_i represents [9].

To solve the global optimization problem, FL relies on local Stochastic Gradient Descent (SGD) to update model parameters on each client. Instead of performing a single gradient step, each client typically executes multiple local SGD iterations on mini-batches from its dataset before communicating with the server. This process can be described as:

$$w_k^{(t,s+1)} \leftarrow w_k^{(t,s)} - \eta \nabla F_k \left(w_k^{(t,s)} \right), \tag{3}$$

where η is the learning rate, and $\nabla F_k(w_k^{(t,s)})$ is the gradient of the local loss function computed on a mini-batch at step s. These local updates accumulate during the local training phase, starting as $w_k^{(t,0)}$. After a predefined number of local steps, denoted E, each client obtains its final local model $w_k^{(t,E)}$, which is then sent to the global server. The global server aggregates the models to form a global model. This can be done using orthogonal FL, or Over-the-Air FL [2].

C. Orthogonal Federated Learning

Orthogonal FL utilizes orthogonal communication and FedAvg. FedAvg is a widely adopted learning algorithm in FL. It defines how local model updates from multiple clients are combined at the server, independent of the communication strategy used to transmit those updates. Orthogonal FL operates iteratively over multiple communication rounds. In each round, a randomly selected subset of the clients, or all clients, train their local models using Eq. (3). Upon completing local training, each client sends its updated model $w_k^{(t,E)}$ to the server, which aggregates them using FedAvg:

$$w_{\text{global}}^{(t)} = \sum_{k=1}^{K} \frac{n_k}{n} w_k^{(t,E)}, \tag{4}$$

where n_k is the number of data points on client k, and $n = \sum_{k=1}^{K} n_k$ is the total number of samples across all participating clients. This averaging ensures that clients with more data have a proportionally greater influence on the global model. The new global model $w_{\text{global}}^{(t)}$ is then distributed back to all clients, where it is used as the starting point for the next round of local updates, in other words $w_k^{(t+1,0)} = w_{\text{global}}^{(t)}$, through SGD:

$$w_k^{(t+1,s+1)} \leftarrow w_k^{(t+1,s)} - \eta \nabla F_k \left(w_k^{(t+1,s)} \right) \tag{5}$$

[10].

A key strength of orthogonal FL is its idealized communication structure. This controlled aggregation process contributes to stable convergence and enables the global model to achieve high accuracy across rounds. However, the synchronized nature of orthogonal FL also introduces communication overhead, as all participating clients transmit sequentially. Beyond latency, orthogonal communication also suffers from poor bandwidth efficiency. Each of the K clients requires a dedicated bandwidth B, leading to a total bandwidth consumption of $K \times B$ [10].

D. Over-the-Air Federated Learning

To address the inefficiencies of orthogonal communication, AirComp can be adopted as a resource-efficient approximation of FedAvg aggregation using non-orthogonal communication. AirComp enables the computation of nomographic functions, such as weighted averages, directly over a wireless multiple access channel (MAC) [11]. AirComp leverages the superposition property of wireless signals to perform aggregation during transmission, allowing multiple clients to transmit their updates simultaneously. This significantly reduces communication latency and eliminates the need for sequential transmissions. The integration of AirComp with FL is commonly referred to as Over-the-Air FL. In contrast to orthogonal FL, which requires a total bandwidth of $K \times B$, Over-the-Air FL performs aggregation using only a single shared bandwidth B, regardless of the number of participating clients. This is made possible by AirComp's concurrent analog transmissions [10].

As in orthogonal FL, each client first performs several iterations of SGD on its local data to obtain an updated model $w_k^{(t,E)}$, which is then transmitted using the AirComp protocol. In the ideal case of Over-the-Air FL, all clients participate in the aggregation process. Each client transmits a scaled version of its local model update $w_k^{(t,E)}$ using a power control coefficient p_k. The transmission is given by:

$$x_k = p_k w_k^{(t,E)}, \tag{6}$$

where ideal channel inversion is applied such that $p_k = \frac{1}{h_k}$, and $h_k \sim \mathcal{CN}(0,1)$ denotes the complex Rayleigh fading channel coefficient of client k. This coefficient models the effect of multipath fading in wireless communication, where the transmitted signal reaches the receiver through multiple paths with varying delays and phases, resulting in random amplitude and phase variations. The received signal at the server becomes:

$$y = \sum_{k=1}^{K} h_k x_k + \xi = \sum_{k=1}^{K} w_k^{(t,E)} + \xi, \tag{7}$$

where $\xi \sim \mathcal{N}(0, \sigma^2)$ represents additive white Gaussian noise (AWGN). The global model is obtained by averaging:

$$w_{\text{global}}^{(t)} = \frac{1}{K} \left(\sum_{k=1}^{K} w_k^{(t,E)} + \xi \right). \tag{8}$$

If the noise is neglected, Eq. (4) is essentially recreated. However, a major practical limitation of this ideal channel inversion is the potentially excessive transmission power required when the channel gain h_k is small. Since ideal channel inversion involves scaling the transmitted signal by $\frac{1}{h_k}$, a very small h_k would require a very large scaling factor. This could result in transmission powers that exceed the physical capabilities of the client device [10].

To counteract this, not all clients are active in every communication round. To reflect a more realistic scenario, clients are selected based on whether their instantaneous channel gain satisfies $|h_k|^2 \geq \gamma$, where γ is a predefined threshold ensuring a practical transmission power. The received signal in this threshold-based Over-the-Air FL model becomes:

$$y = \sum_{k=1}^{K} h_k p_k w_k^{(t,E)} + \xi, \tag{9}$$

with the individual power control coefficient defined as:

$$p_k = \begin{cases} \frac{\sqrt{\rho}}{h_k}, & \text{if } |h_k|^2 \geq \gamma, \\ 0, & \text{otherwise}, \end{cases} \tag{10}$$

where ρ is a global scaling parameter. This leads to the simplified received signal:

$$y = \sum_{k \in A} \sqrt{\rho} w_k^{(t,E)} + \xi, \tag{11}$$

where A is the set of active clients satisfying the threshold condition. The server then normalizes the result:

$$w_{\text{global}}^{(t)} = \frac{1}{|A|} \sum_{k \in A} w_k^{(t,E)} + \frac{\xi}{|A|\sqrt{\rho}}. \tag{12}$$

Clients continue to train using the global model, that is, $w_k^{(t+1,0)} = w_{\text{global}}^{(t)}$, through SGD as in Eq. (5) [10].

To minimize the effect of noise, the parameter ρ should be as large as possible under a power constraint. Each client must satisfy:

$$E[|p_k|^2] \leq P_0, \tag{13}$$

where P_0 is the maximum allowable average transmit power per client. Substituting the definition of $p_k = \sqrt{\rho}/h_k$, the expected transmit power becomes:

$$E[|p_k|^2] = \rho \cdot E\left[\frac{1}{|h_k|^2} \,\middle|\, |h_k|^2 \geq \gamma \right]. \tag{14}$$

Under Rayleigh fading, where $|h_k|^2 \sim \text{Exp}(1)$, the conditional expectation of the inverse channel gain is given by:

$$E\left[\frac{1}{|h_k|^2} \,\middle|\, |h_k|^2 \geq \gamma \right] = \int_{\gamma}^{\infty} \frac{e^{-t}}{t}\, dt = \text{Ei}(\gamma), \tag{15}$$

where $\text{Ei}(\gamma)$ is the exponential integral function. The exponential integral arises naturally in this context, as it captures the average transmission energy required when only clients with channel gains above a threshold are allowed to transmit. It reflects the cost of inverting the channel over a truncated exponential distribution. Substituting into the power constraint:

$$\rho \cdot \text{Ei}(\gamma) \leq P_0, \tag{16}$$

which yields the upper bound:

$$\rho \leq \frac{P_0}{\text{Ei}(\gamma)}. \tag{17}$$

This ensures that the expected transmission energy remains within the allowed budget while maintaining robustness against channel fading [10].

Over-the-Air FL has emerged as a promising technique for reducing communication latency and enhancing scalability in FL by enabling simultaneous analog transmission from multiple clients [10]. This makes it particularly attractive for large-scale deployments where communication bottlenecks are critical. However, several practical challenges remain. These include noise accumulation due to the superposition nature of the channel, the necessity of precise power control for accurate signal alignment, and strict requirements on timing synchronization, all of which can adversely affect model performance and system stability [4], [12].

To address these limitations, threshold-based client selection can be employed, where only clients with sufficiently strong channel conditions are allowed to participate in model aggregation [12], [13]. This strategy offers a trade-off between communication efficiency and model accuracy by reducing the impact of poor-quality channels and mitigating excessive noise. As a result, it improves the reliability of Over-the-Air FL in realistic wireless environments while still preserving its inherent benefits of low latency and scalability [14].

III. Method

This section outlines the specific implementation details used to simulate and evaluate orthogonal Federated Learning (FL) and Over-the-Air FL. It begins with a description of the neural network architectures, followed by the preprocessing of datasets and the design of data distributions. The section then presents the FL setup for both paradigms and explains how the two communication methods were evaluated.

A. Neural Network Architectures and Data Processing

Two neural network models were implemented to accommodate the complexity of the datasets: a Multilayer Perceptron (MLP) for MNIST and a Convolutional Neural Network (CNN) for CIFAR-10.

The architecture of the MLP used for image classification is summarized in Table I. This simple feedforward network is composed of two fully connected layers with one activation function in between. The input image is first flattened from its original shape, 28×28 pixels, into a 784-dimensional vector. This vector is then passed through a fully connected layer with 128 neurons, which transforms the input into a lower-dimensional feature representation. A Rectified Linear Unit (ReLU) activation function follows, introducing non-linearity by setting all negative values to zero and leaving positive values unchanged. Finally, the output layer maps the 128-dimensional features to 10 logits, corresponding to the 10 possible classes in the classification task.

TABLE I
MLP Architecture

Layer	Configuration
Input	Flatten 28×28 to 784
Fully Connected 1	Linear(784, 128)
Activation	ReLU
Fully Connected 2	Linear(128, 10)
Output	Logits for 10 classes

The CNN model used in this study has a moderately deep architecture with two convolutional blocks, as detailed in Table II. It is designed to process 32×32 RGB images, each composed of three channels representing red, green, and blue components. In the first convolutional block, the model applies a two-dimensional convolutional layer with 3×3 kernels and padding to maintain spatial dimensions. Batch normalization (BN) is applied to stabilize learning, followed by a ReLU activation to introduce non-linearity. Max pooling with a 2×2 kernel is then used to reduce the resolution, helping the model focus on salient features. The second block repeats this structure but increases the number of output channels, enabling the model to extract more complex features. After two pooling operations, the spatial resolution is reduced from

32×32 to 8×8. The resulting feature maps are flattened into a 4096-dimensional vector and passed through a fully connected layer that outputs 10 logits, corresponding to the classification categories.

TABLE II
CNN ARCHITECTURE

Layer	Configuration
Input	32×32 RGB image
Conv Block 1	Conv2D(3, 32, kernel=3, padding=1) + BN + ReLU + MaxPool(2×2)
Conv Block 2	Conv2D(32, 64, kernel=3, padding=1) + BN + ReLU + MaxPool(2×2)
Flatten	64×8×8 = 4096
Fully Connected	Linear(4096, 10)
Output	Logits for 10 classes

For MNIST, the grayscale images were converted to tensors and normalized to the $[0, 1]$ range to ensure numerical stability and faster convergence during training. For CIFAR-10, a more comprehensive transformation pipeline was used for the data, including random horizontal flipping, random rotations, and normalization with a mean and standard deviation of 0.5 per channel.

To simulate realistic data distribution scenarios in an FL context, MNIST was partitioned between clients using two distinct strategies: Independent and Identically Distributed (IID) and non-Independent and Identically Distributed (non-IID). In the IID setting, data samples were randomly and evenly distributed among clients, ensuring that each client received a balanced representation of all classes. In the non-IID setting, data was partitioned based on class labels, such that each client received samples biased toward only two specific classes. For instance, one client could receive samples exclusively from digits zero and one, while another client would receive samples from digits six and nine. This form of label-based partitioning introduced significant statistical heterogeneity across clients and allowed for evaluating model robustness under more realistic and challenging conditions, where local data distributions vary substantially.

B. Federated Learning Setup

A FL environment was simulated with one central server and multiple clients. Both orthogonal FL and Over-the-Air FL were implemented.

Orthogonal Federated Learning Setup: In the orthogonal FL setup, global model updates were processed with Federated Averaging (FedAvg) in the simulation to emulate aggregation without interference. Each client received a copy of the global model and performed local training using Stochastic Gradient Descent (SGD). During the local training phase, each client applied multiple SGD updates by computing gradients over mini-batches from its local dataset according to Eq. (3). After training, clients returned their updated model parameters to the central server.

The server aggregated the received model updates using weighted averaging according to Eq. (4), where each client's contribution was proportional to the size of its local dataset. This ensured that clients with more data had a larger influence on the global model. The aggregation process was repeated over multiple communication rounds, allowing the global model to be iteratively refined.

Over-the-Air Federated Learning Setup: In the Over-the-Air FL simulation, the superposition property of a wireless medium was mathematically modeled to emulate in-air aggregation of model updates. Each client's local model was trained with SGD and scaled based on a simulated channel gain according to Eq. 10, to reflect channel inversion. The wireless environment was simulated using Rayleigh fading and additive white Gaussian noise (AWGN) to introduce randomness and noise into the aggregation process, so the combined transmission is given by Eq. (9). Accurate channel state information was assumed to be available at each client to correctly scale transmissions within the simulation. Channel state information (CSI) refers to the knowledge of the channel's current fading conditions, including amplitude and phase, which is essential for adjusting transmission parameters such as power and phase to achieve coherent signal aggregation.

At the server side of the simulation, the aggregated signal was represented as a normalized noisy superposition of the clients' model updates as seen in Eq. (12). This aggregated model was then used to update the global model iteratively over several communication rounds. Although no actual transmission occurred, the simulation captured the key characteristics of Over-the-Air FL, including its reliance on simultaneous aggregation and sensitivity to noise. The reliability of the aggregation process was quantified using the exponential integral in Eq. (15).

C. Hyperparameter Tuning and Evaluation Strategy

The hyperparameters were either fixed to specific values across all simulations or varied within a defined interval for tuning purposes. The fixed values were chosen based on preliminary manual experimentation, during which each parameter was systematically varied across multiple simulation runs to identify the most effective configuration, balancing the trade-off between model accuracy and execution time. One or two optimal values, depending on the simulation scenario, were similarly identified for each varying variable, allowing it to be fixed during specific simulations. The hyperparameters used in the orthogonal FL setup are summarized in Table III.

TABLE III
ORTHOGONAL FL HYPERPARAMETER SETTINGS

Parameter	Fixed Value	Variable Interval
Number of clients K	–	20–50
Number of rounds t	100	–
Local epochs	–	1–5
Batch size	64	–
Learning rate η	0.01	–

The Over-the-Air FL setup involves additional hyperparameters compared to orthogonal FL, as summarized in Table IV.

The evaluation of the two communication strategies, orthogonal FL and Over-the-Air FL, was designed to pinpoint the impact of one parameter at a time. All simulations were conducted with both orthogonal FL and Over-the-Air FL over 100 communication rounds using consistent settings with a fixed learning rate of $\eta = 0.01$. The simulations focused on five key aspects: client participation, number of local training

TABLE IV
OVER-THE-AIR FL HYPERPARAMETER SETTINGS

Parameter	Fixed Value	Variable Interval
Number of clients K	–	20–50
Number of rounds t	100	–
Local epochs	–	1–5
Batch size	64	–
Learning rate η	0.01	–
Noise variance σ^2	–	10^{-7}–10^{-2}
Threshold γ	–	0.1–0.2
Average transmit power constraint P_0	–	0.2–0.5

epochs, dataset complexity, data distribution, and the Over-the-Air FL-specific parameter noise variance.

To investigate the influence of client participation on model convergence, simulations were conducted using the MNIST dataset under an IID data distribution. Experiments were performed with one local epoch per round. The communication parameters for Over-the-Air FL were consistently set with a noise variance of $\sigma^2 = 10^{-3}$, a threshold $\gamma = 0.2$, and a transmission power $P_0 = 0.2$. To further explore the effect of local computation intensity, additional simulations varied the number of local epochs per client while maintaining the number of participating clients constant at 20. The other parameters remained the same as in the client participation simulation.

To examine the influence of dataset complexity, simulations were conducted on both MNIST and CIFAR-10 using 20 clients and two local epochs per round. Over-the-Air FL parameters were set to a noise variance of $\sigma^2 = 10^{-7}$, threshold $\gamma = 0.1$, and transmission power $P_0 = 0.5$. This setup enabled a comparison of how each method handles increased data dimensionality and visual complexity. To investigate the impact of data heterogeneity, additional simulations were performed on the MNIST dataset under both IID and non-IID distributions. In this case, the number of clients remained at 20, while the number of local epochs was reduced to one. The communication parameters were adjusted to $\sigma^2 = 10^{-3}$, $\gamma = 0.2$, and $P_0 = 0.2$. Lastly, to evaluate the robustness of Over-the-Air FL to communication noise, simulations were performed with varying noise variance σ^2 while maintaining 20 clients, one local epoch, $\gamma = 0.2$, and $P_0 = 0.2$.

This systematic approach enabled both an intra-strategy analysis, where each method was evaluated across multiple controlled variations, and an inter-strategy comparison, as the same experimental configurations were used to compare orthogonal FL and Over-the-Air FL under equivalent conditions.

D. Tools and Environment

The entire implementation was developed in Python using several key libraries. PyTorch was used to build and train neural network models and to perform gradient-based optimization. The torchvision library enabled access to standard image datasets and preprocessing tools. NumPy was used for handling arrays and numerical computations, while Matplotlib was used to visualize the results. SciPy was used for advanced mathematical operations, including the exponential integral function demonstrated in Eq. (15). All code was written and run locally in Visual Studio Code without GPU acceleration.

IV. RESULTS

This section presents the experimental results obtained from simulations of Federated Learning (FL) conducted under both orthogonal and non-orthogonal communication scenarios. The first part presents the impact of varying the number of participating clients and the number of local training epochs on model convergence for both orthogonal FL and Over-the-Air FL. This is followed by how different datasets and data distributions influence convergence in both communication settings. Finally, the effect of noise variance in Over-the-Air FL is presented.

A. Client Participation

The impact of the number of participating clients on model convergence in FL is illustrated in Fig. 2 for orthogonal FL and in Fig. 3 for Over-the-Air FL, both evaluated over 100 communication rounds. The simulations were performed using the MNIST dataset with Independent and Identically Distributed (IID) data. Each client conducted one local training epoch per round and with a learning rate η of 0.01. The communication parameters for Over-the-Air FL were consistently set with a noise variance of $\sigma^2 = 10^{-3}$, a threshold $\gamma = 0.2$, and a transmission power $P_0 = 0.2$. A Multilayer Perceptron (MLP) was used.

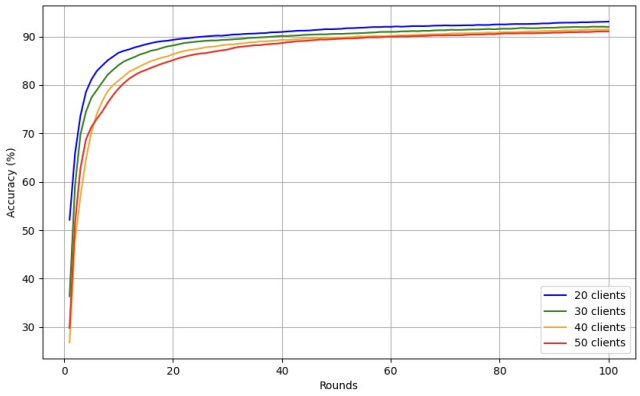


Fig. 2. Orthogonal FL with a varying number of participating clients between 20–50.

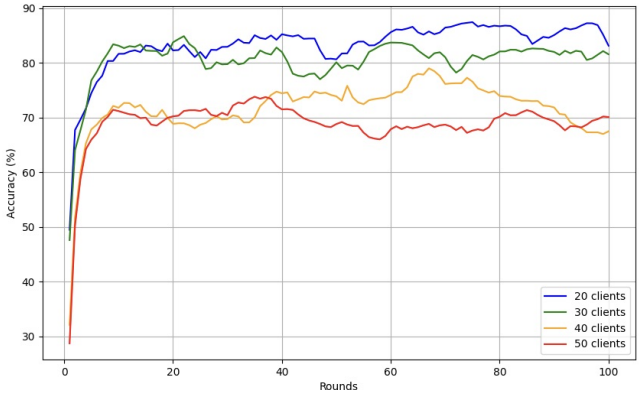


Fig. 3. Over-the-Air FL with a varying number of participating clients between 20–50.

B. Local Training Epochs

The effect of local epochs on model convergence in FL is shown in Fig.4 for orthogonal FL and in Fig.5 for Over-the-Air FL, respectively, as the number of local training epochs per client is varied. Each simulation was conducted over 100 communication rounds using the MNIST dataset under an IID data distribution. The number of participating clients was fixed at 20, and all models were trained with a learning rate η of 0.01. The Over-the-Air FL parameters were consistently set with a noise variance of $\sigma^2 = 10^{-3}$, a threshold $\gamma = 0.2$, and a transmission power $P_0 = 0.2$. Again, the MLP architecture was used.

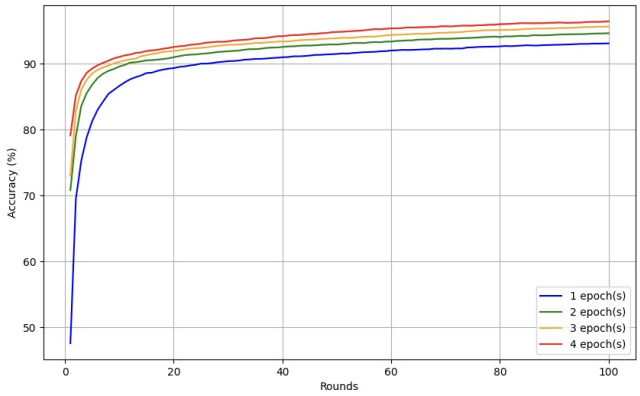


Fig. 4. Orthogonal FL with a varying number of local training epochs between 1–4.

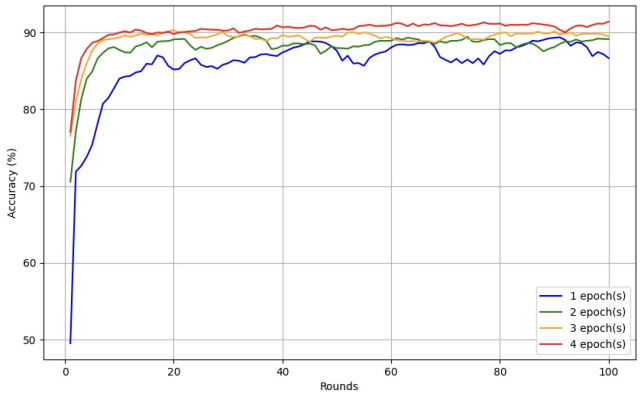


Fig. 5. Over-the-Air FL with a varying number of local training epochs between 1–4.

C. Dataset Complexity

This section illustrates how dataset complexity influences convergence behavior. The evaluation compares performance on the MNIST dataset, using an MLP, and the CIFAR-10 dataset, using a Convolutional Neural Network (CNN), under an IID data distribution. Each simulation was carried out for 100 communication rounds with 20 clients, a fixed learning rate of $\eta = 0.01$, and two local epochs per round. The Over-the-Air FL parameters were set to a noise variance of $\sigma^2 = 10^{-7}$, a threshold $\gamma = 0.1$, and a transmission power $P_0 = 0.5$. Fig. 6 and Fig. 7 show the resulting accuracy curves for orthogonal FL and Over-the-Air FL, respectively.

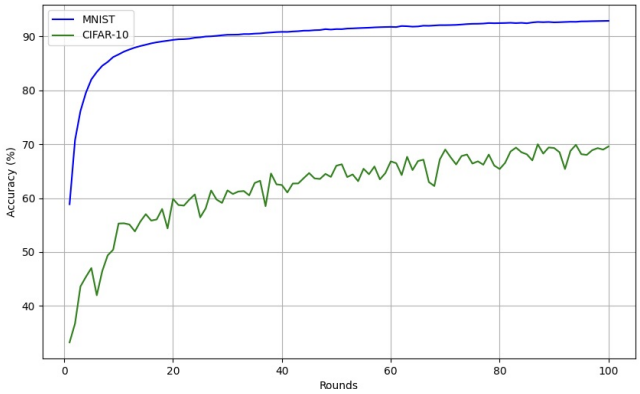


Fig. 6. Orthogonal FL with varying dataset complexity: MNIST versus CIFAR-10.

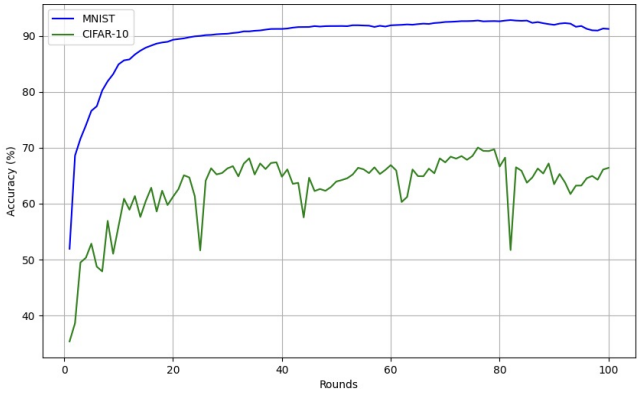


Fig. 7. Over-the-Air FL with varying dataset complexity: MNIST versus CIFAR-10.

D. Data Distribution

To illustrate the sensitivity of FL methods to data heterogeneity, the MNIST dataset was used under both IID and non-Independent and Identically Distributed (non-IID) configurations. The simulations were performed using an MLP and identical settings: 100 communication rounds, 20 clients, a learning rate of $\eta = 0.01$, and one local training epoch per round. For the Over-the-Air FL-based approach, a noise variance of $\sigma^2 = 10^{-3}$ was applied, along with transmission parameters $P_0 = 0.2$ and $\gamma = 0.2$. Fig. 8 and Fig. 9 show how the change in data distribution influences model convergence for orthogonal FL and Over-the-Air FL, respectively.

E. Over-the-Air Federated Learning Specific Parameters

To investigate the robustness of Over-the-Air FL to channel noise, simulations were conducted on the MNIST dataset under an IID data distribution while varying the noise variance σ^2. All other parameters were kept constant across runs: 100 communication rounds, 20 clients, one local training epoch per round, and a learning rate of $\eta = 0.01$. Transmission settings were fixed at $P_0 = 0.2$ and $\gamma = 0.2$. Fig. 10 displays how different levels of noise variance impact the convergence. The MLP architecture was used.

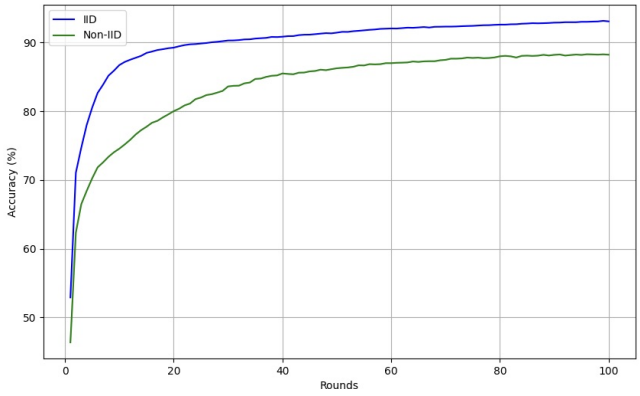


Fig. 8. Orthogonal FL with varying data distribution: IID versus non-IID.

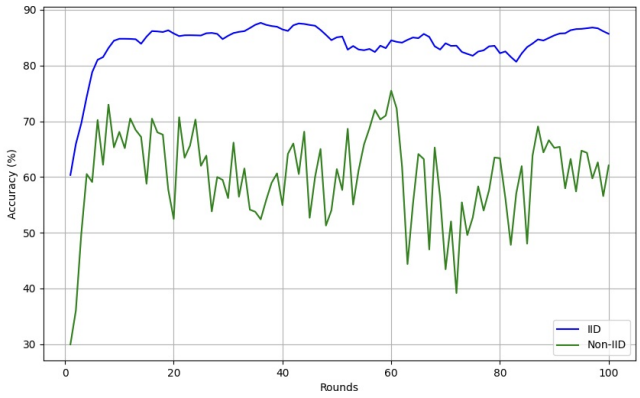


Fig. 9. Over-the-Air FL with varying data distribution: IID versus non-IID.

V. Discussion

This section provides an interpretation of the experimental results presented above, highlighting key insights and implications related to the performance and communication efficiency of orthogonal Federated Learning (FL) and Over-the-Air FL.

A. Client Participation

Client participation in orthogonal FL has a direct impact on both the overall model accuracy and the convergence speed, as shown in Fig. 2. It is evident that a lower number of clients results in both higher final accuracy and faster convergence.

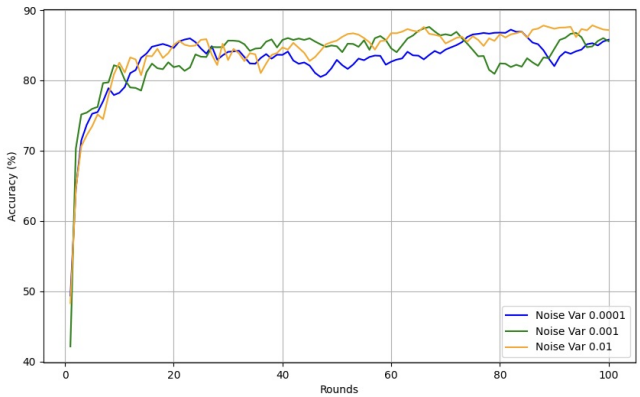


Fig. 10. Over-the-Air FL with varying noise variance between 10^{-4}–10^{-2}.

A similar pattern can be observed for Over-the-Air FL in Fig. 3, where fewer clients also lead to better performance, although the effect is less pronounced than in orthogonal FL. This is likely due to the influence of noise and the stochastic nature of client participation in Over-the-Air FL.

Overall, a lower number of participating clients appears to improve model accuracy in both orthogonal FL and Over-the-Air FL. This can be attributed to more consistent local updates, as each client trains on a larger share of the data. Over-the-Air FL consistently achieves significantly lower accuracy across all levels of client participation, highlighting the advantages of orthogonal communication in this theoretical study. An increased number of clients also leads to higher execution times in both scenarios due to the greater number of local models that need to be trained and aggregated during each communication round.

B. Local Training Epochs

Varying the number of local training epochs in orthogonal FL has a direct impact on both the convergence speed and the final model accuracy, as shown in Fig. 4. A higher number of epochs results in better overall performance, with four local epochs achieving the highest accuracy, followed by three, two, and one. The relationship is consistent across all configurations, indicating that increased local computation leads to more informative updates and faster convergence.

Over-the-Air FL demonstrates a similar overall trend in Fig. 5, where four epochs again lead to the highest accuracy and one epoch performs the worst. However, unlike in orthogonal FL, the convergence curves for different epoch settings exhibit small but consistent overlaps throughout training. This suggests that while more local training still improves performance, the stochastic effects of noise and variable client participation in Over-the-Air FL introduce fluctuations that can momentarily mask the benefits of additional epochs. As a result, the positive effect of increased local training is still present, but less stable and more sensitive to random variation in each round.

Increasing the number of local epochs improves performance in both orthogonal FL and Over-the-Air FL, with both methods reaching comparable accuracy levels across different epoch settings. However, the improvements are more consistent and predictable in orthogonal FL, where the deterministic and noise-free aggregation ensures that stronger local training directly translates into better global updates. In contrast, Over-the-Air FL's performance gains are partially offset where noise and inconsistent participation can disrupt the expected benefits of local training. These results underline the robustness of orthogonal FL in scenarios with reliable communication and highlight the trade-offs introduced by Over-the-Air FL's communication efficiency at the cost of precision.

C. Dataset Complexity

The impact of dataset complexity between MNIST and CIFAR-10 is observed in the orthogonal FL simulations, where the model achieves significantly higher performance on MNIST, as shown in Fig. 6. The lower and more unstable

results are expected for simulations with CIFAR-10, as the complexity of the dataset makes the learning task more challenging for the model.

This pattern is also evident in Over-the-Air FL, as shown in Fig. 7, where MNIST converges stably and rapidly, while CIFAR-10 remains unstable and at significantly lower levels of accuracy. This behavior can be attributed to the combination of high dataset complexity and the inherent noise sensitivity of Over-the-Air FL aggregation.

It is important to note that the observed differences between simulations on MNIST and CIFAR-10 within both orthogonal FL and Over-the-Air FL would likely have been even more pronounced if the same neural network architecture had been used for both datasets. In the current setup, a Multilayer Perceptron (MLP) is employed for MNIST, while a more powerful Convolutional Neural Network (CNN) is used for CIFAR-10 to better accommodate the dataset's higher complexity. This architectural adjustment for CIFAR-10 helps mitigate the challenges posed by its increased dimensionality and visual complexity, thereby narrowing the performance gap between the two datasets. Nevertheless, the superior performance observed on MNIST can primarily be attributed to its simpler structure, which facilitates faster and more stable model convergence.

Higher dataset complexity lowers both the accuracy and convergence stability of orthogonal FL and Over-the-Air FL; however, Over-the-Air FL is more sensitive to increases in data dimensionality due to its unstable nature. It is worth noting that the performance gap between orthogonal FL and Over-the-Air FL is relatively small in this case. This suggests that dataset complexity does not have a significantly different impact across the two communication settings.

D. Data Distribution

Fig. 8 shows that orthogonal FL achieves better performance under Independent and Identically Distributed (IID) conditions compared to non-Independent and Identically Distributed (non-IID). The IID setting allows the model to converge rapidly, reaching over 90% accuracy within the first 30 rounds. In contrast, the non-IID scenario converges more slowly and plateaus at a lower accuracy, around 89%. This performance gap is attributed to statistical heterogeneity in the non-IID case, where local updates are biased and less representative of the global objective, making it harder for the model to generalize.

A similar but more pronounced effect is observed in Over-the-Air FL, as shown in Fig. 9. While the IID setting yields relatively stable convergence near 90% accuracy, the non-IID case suffers from significant fluctuations throughout the simulation. These oscillations highlight the difficulty of aggregating non-uniform updates over a noisy analog channel. Although the non-IID setting in the Over-the-Air FL model at one point approaches approximately 75% accuracy, it does so with considerable instability, likely due to the compounded effects of both data heterogeneity and transmission noise.

In summary, both orthogonal FL and Over-the-Air FL benefit from IID data distributions, but the impact is more severe in the latter. Orthogonal FL shows relatively stable convergence under non-IID settings, whereas Over-the-Air FL struggles with high fluctuations and poor convergence. This demonstrates the combined challenge of statistical heterogeneity and analog aggregation noise in Over-the-Air FL, emphasizing the importance of designing robust algorithms or data preprocessing techniques when operating under non-ideal data distributions.

E. Over-the-Air Federated Learning Specific Parameters

Noise variance was varied to evaluate the robustness of Over-the-Air FL. As expected, higher noise levels degrade model performance. When the noise variance was low, Over-the-Air FL provided a slightly smoother accuracy curve. These results highlight the sensitivity to channel noise in Over-the-Air FL as shown in Fig. 10.

The best overall convergence in the Over-the-Air FL simulations was observed with a threshold of $\gamma = 0.2$, a power constraint of $P_0 = 0.2$, and a noise variance of $\sigma^2 = 10^{-3}$. While similar values of the power constraint and the threshold generally led to the most stable performance, simulations involving more complex datasets showed slightly improved convergence with lower threshold values and higher power constraints. This may be because more complex datasets require higher signal quality during aggregation, which can be achieved by lowering the threshold and increasing the transmission power. Lower values of noise were also beneficial in more complex simulations, as noise accumulation became more evident.

F. Cross-Parameter Dynamics

While each parameter was evaluated individually, several trends emerged when considering how the different parameters compare in magnitude and stability of their effects. Most notably, client participation and local training epochs exhibit contrasting dynamics. Increasing the number of clients generally results in lower final accuracy and slower convergence, whereas increasing the number of local epochs consistently improves both metrics. This trade-off highlights a tension between broader data representation and update consistency. With more clients, each contributes less to the global model, and the variance in updates increases. In contrast, more local training allows each client to extract more meaningful gradients before aggregation, resulting in more stable global updates. Interestingly, the impact of increasing clients is more severe than decreasing the number of local epochs, particularly in settings where convergence stability is critical. For instance, the difference in accuracy between 20 and 50 clients is significantly larger than the difference between one and four local epochs. This suggests that client participation is a more sensitive parameter to tune in practice, especially when dealing with constrained communication or noisy aggregation.

When comparing the effects of dataset complexity and data distribution, a clear distinction emerges in how each factor influences training stability and final performance. Increasing dataset complexity, by moving from MNIST to CIFAR-10, leads to a gradual and consistent drop in accuracy, accompanied by fluctuations. This suggests that more complex datasets

make the learning task inherently harder, but in a relatively predictable manner, as the convergence is still visible. In contrast, changing the data distribution from IID to non-IID introduces significantly more unpredictability for the non-orthogonal case than the orthogonal case. The accuracy curves under non-IID conditions converge irregularly. This volatility suggests that data heterogeneity affects not only the final accuracy but also the entire learning trajectory, making it more difficult to interpret or trust the model's convergence.

Finally, noise variance shows a moderate but non-negligible effect, primarily interacting with other parameters such as client count and data distribution. While not the most critical factor in isolation, its compounding effects can exacerbate issues introduced by other variables.

In summary, client participation and local training epochs show opposing effects: more clients lead to lower accuracy and slower convergence, while more local epochs improve both. Dataset complexity results in a predictable performance drop, whereas non-IID distributions introduce instability, particularly for Over-the-Air FL. Lastly, noise variance has a moderate but compounding impact, amplifying the negative effects of other parameters such as high client count and heterogeneous data.

G. Communication Resource Efficiency

Beyond accuracy and convergence behavior, an important distinction between orthogonal FL and Over-the-Air FL lies in their communication resource requirements. In orthogonal FL, orthogonal communication implies that each client transmits its model update over a separate bandwidth B or time slot. This results in a linear increase in communication resource consumption with the number of clients, which can significantly limit scalability in bandwidth- or time-constrained systems. Since each client requires a separate communication resource, in the case of bandwidth, the total bandwidth demand scales proportionally with the number of clients K, resulting in a total consumption of $K \times B$.

In contrast, Over-the-Air FL leverages simultaneous analog aggregation, where all clients transmit their updates concurrently over a shared wireless channel. This decouples the total bandwidth requirement from the number of participating clients, maintaining a constant demand of only B regardless of clients K. The ability to aggregate updates in a single communication step not only minimizes latency but also drastically reduces spectral consumption. As a result, Over-the-Air FL offers superior communication efficiency, making it especially attractive for large-scale deployments where communication resources are limited or costly.

However, this efficiency comes at the cost of increased sensitivity to channel noise and synchronization errors. Because the aggregation is performed directly in the analog domain, noise introduced during transmission cannot be corrected post hoc, leading to direct degradation of the model update. Additionally, Over-the-Air FL relies on precise synchronization and accurate power control, which are difficult to guarantee in practical wireless networks due to channel fading and device heterogeneity. In particular, techniques like channel inversion can cause excessive amplification when channel gains are low, increasing energy usage and sometimes exceeding hardware limits.

Robust aggregation schemes, methods that are not heavily impacted by deviating values, can reduce the sensitivity to noise. Error correction techniques and signal regularization may also offer protection against transmission errors. Furthermore, adaptive client selection, as implemented in this study through thresholding, can exclude clients with poor channel conditions to maintain aggregation quality, although at the cost of reduced participation.

In addition to bandwidth and latency considerations, energy efficiency constitutes another critical factor in the evaluation of communication strategies. Over-the-Air FL has the potential to reduce energy consumption due to its high communication efficiency. This stands in contrast to orthogonal FL, where longer total transmission time tends to increase overall energy usage. However, the energy-saving potential of Over-the-Air FL is highly dependent on the prevailing wireless channel conditions. Power control techniques such as channel inversion can lead to high transmission power when the channel gain is low, also increasing energy consumption. Threshold-based client selection partially alleviates this issue, though at the expense of reduced diversity in the training data. As such, while Over-the-Air FL offers promising energy-saving benefits, they must be carefully balanced against practical limitations in wireless power management and the need for robust, scalable communication protocols.

H. Further Research

Future research should explore the implementation of more realistic wireless communication models to better evaluate the potential advantages of Over-the-Air FL in practical settings. In particular, the ability of Over-the-Air FL to perform simultaneous analog aggregation may offer significant scalability and latency advantages over orthogonal FL in large-scale deployments. Investigating how these benefits manifest under realistic constraints, and when they outweigh the accuracy trade-offs introduced by noise and imperfect power control, remains an important direction for future work.

VI. Conclusion

The primary contribution of this study is an analysis of how various system parameters affect the performance and robustness of both orthogonal Federated Learning (FL) and Over-the-Air FL. Rather than aiming to determine which method is inherently superior, the work focuses on comparing the sensitivity of each approach to critical factors. The study shows that Over-the-Air FL is particularly sensitive to noise and data heterogeneity, with significant degradation in convergence stability and model accuracy. In contrast, orthogonal FL consistently maintains higher accuracy and robustness across all tested configurations, although with greater communication overhead. Despite its lower performance, Over-the-Air FL promises practical advantages in communication efficiency, making it highly relevant for deployment in bandwidth- or

energy-constrained environments. It achieves model aggregation with constant bandwidth regardless of the number of clients, an appealing feature for scalable, real-time applications.

The choice between orthogonal FL and Over-the-Air FL depends on the target application and deployment environment. As FL continues to gain traction as a privacy-preserving paradigm, future research should focus on more realistic wireless environments to enhance the robustness of non-orthogonal communication and better meet the growing demands of decentralized machine learning.

APPENDIX A
GITHUB REPOSITORY

https://github.com/fannyynyberg/
Federated-Learning-and-Over-the-Air-Computation.git

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